



Diploma in

Data Analytics

Lesson 3: Linear regression



Introduction to linear regression



Correlation



Vectors & factors



Lesson Objectives

Lesson 3



Intro to linear regression

Regression analysis

Method for determining the relationship that best fits the observed data



Regression analysis

$$Y = \alpha + \beta + \varepsilon$$

Y = dependant variable

α = intercept

β = slope

ε = random error term



Types of regression

- Linear
- Logistic
- Polynomial
- Lasso
- Ridge
- Random forest





**Did you
know?**



**Legendre and Gauss
issued a paper what is
known as the earliest
form of linear regression
in the early 1900's**



Intro to simple linear regression

- Foundation to more complex and modern techniques
- Models relationship between dependent and independent variable through linear equation
- Least squares method to fit line of best fit





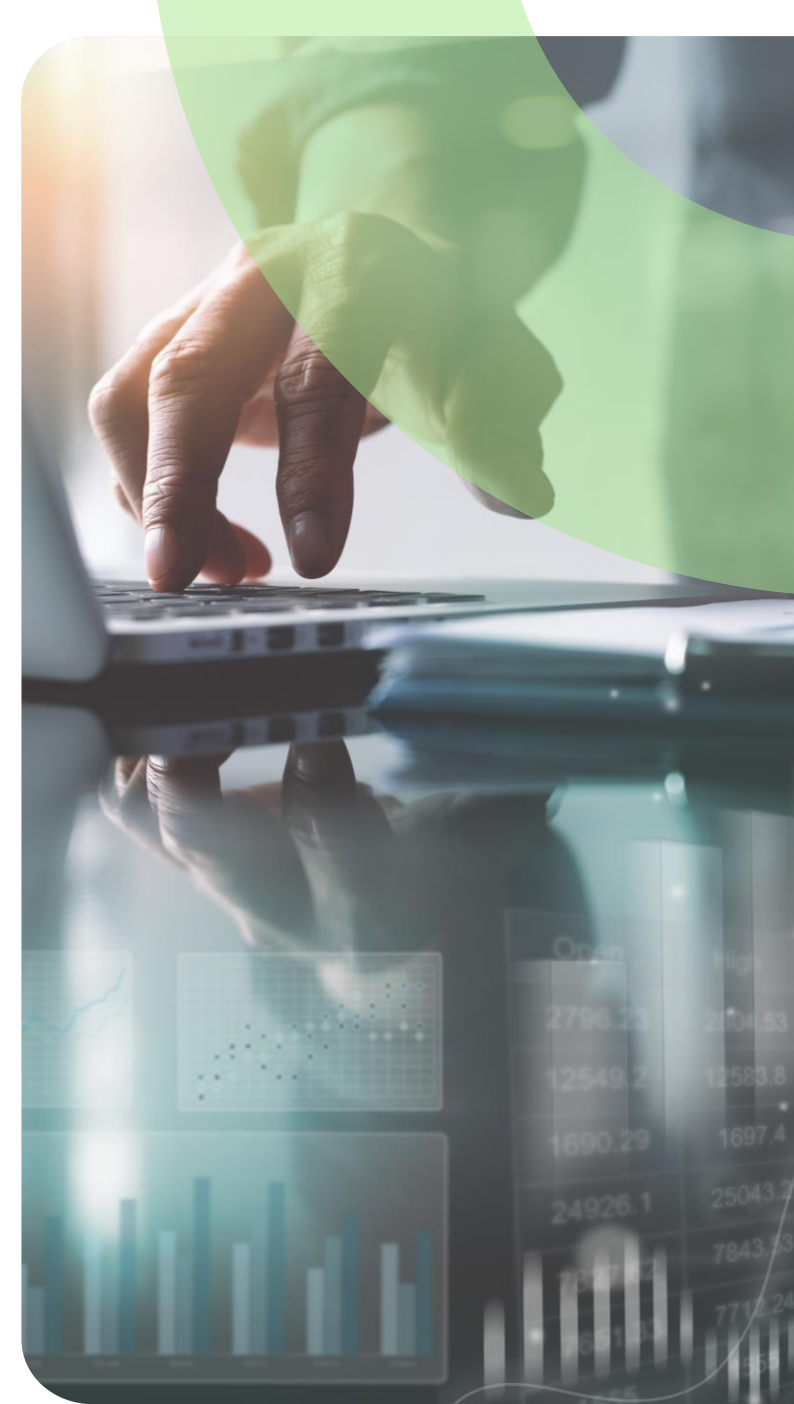
Simple linear regression

$$\hat{y} = a + bx_i$$

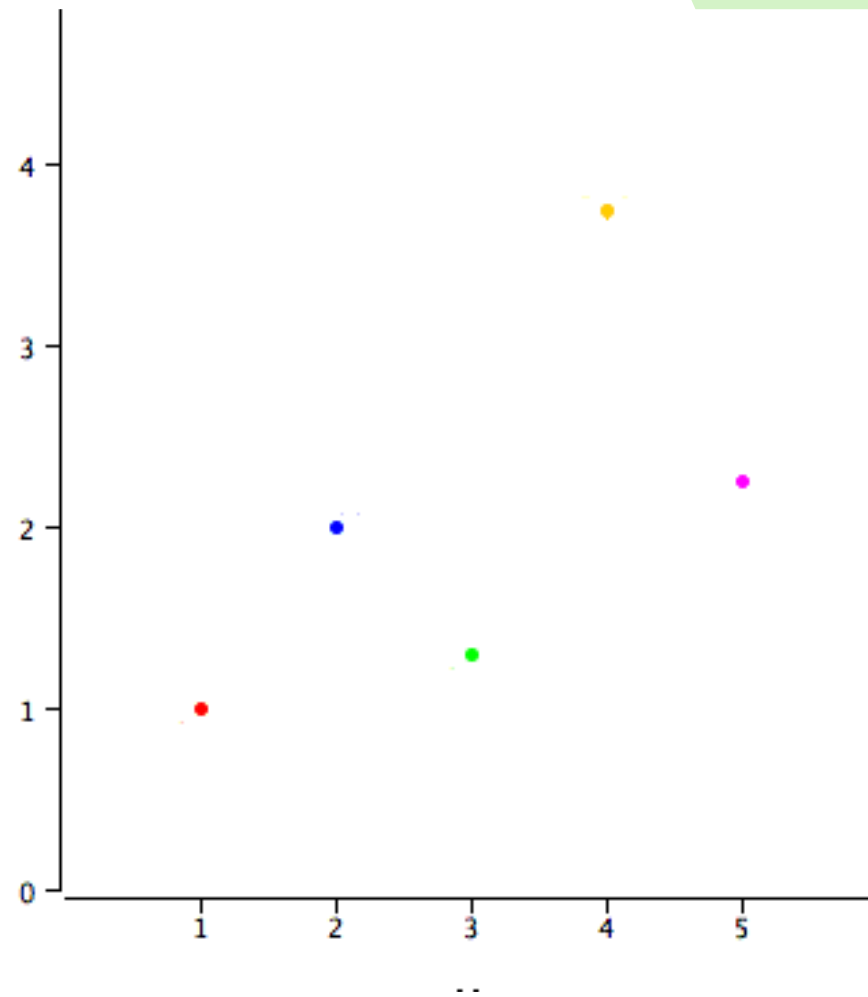
y = fitted or
predicted value

a = estimate of α ,
intercept

b = estimate of β ,
slope

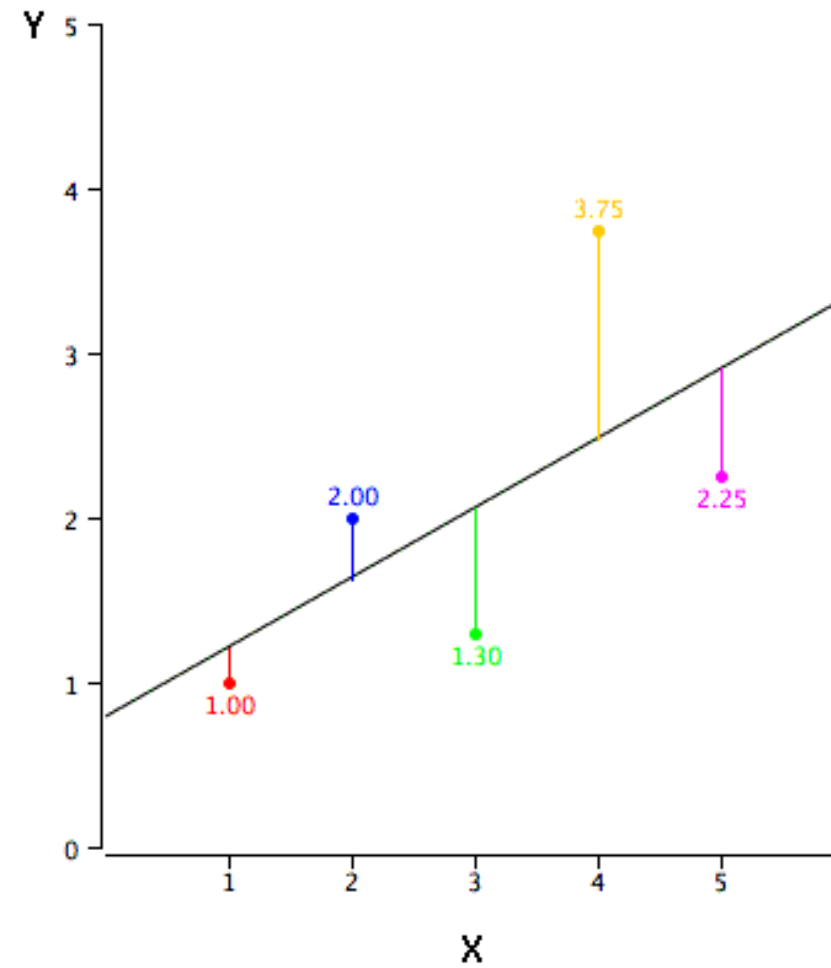


Scatterplot



Source: <http://onlinestatbook.com/2/regression/intro.html>

Line of best fit



Source: <http://onlinestatbook.com/2/regression/intro.html>



Linear regression in R

>lm()

>lm(y ~ x, data)

>lm.fit

> summary (lm.fit)



Correlation

Correlation coefficient

- Measure of association between X and Y in a normal population
- $-1 < \text{corr coeff.} < 1$



Understanding correlation

- Positive correlation
 - Corr coeff. > 0
- Negative correlation
 - Corr coeff. < 0
- No correlation
 - Corr coeff. $= 0$





Population correlation coefficient

$$\rho = \frac{cor(X, Y)}{\sqrt{var(X)var(Y)}}, -1 < \rho < 1$$

Sample correlation coefficient

$$r = \frac{SS_{xy}}{\sqrt{SS_x SS_y}}, -1 < r < 1$$



Correlation vs causation

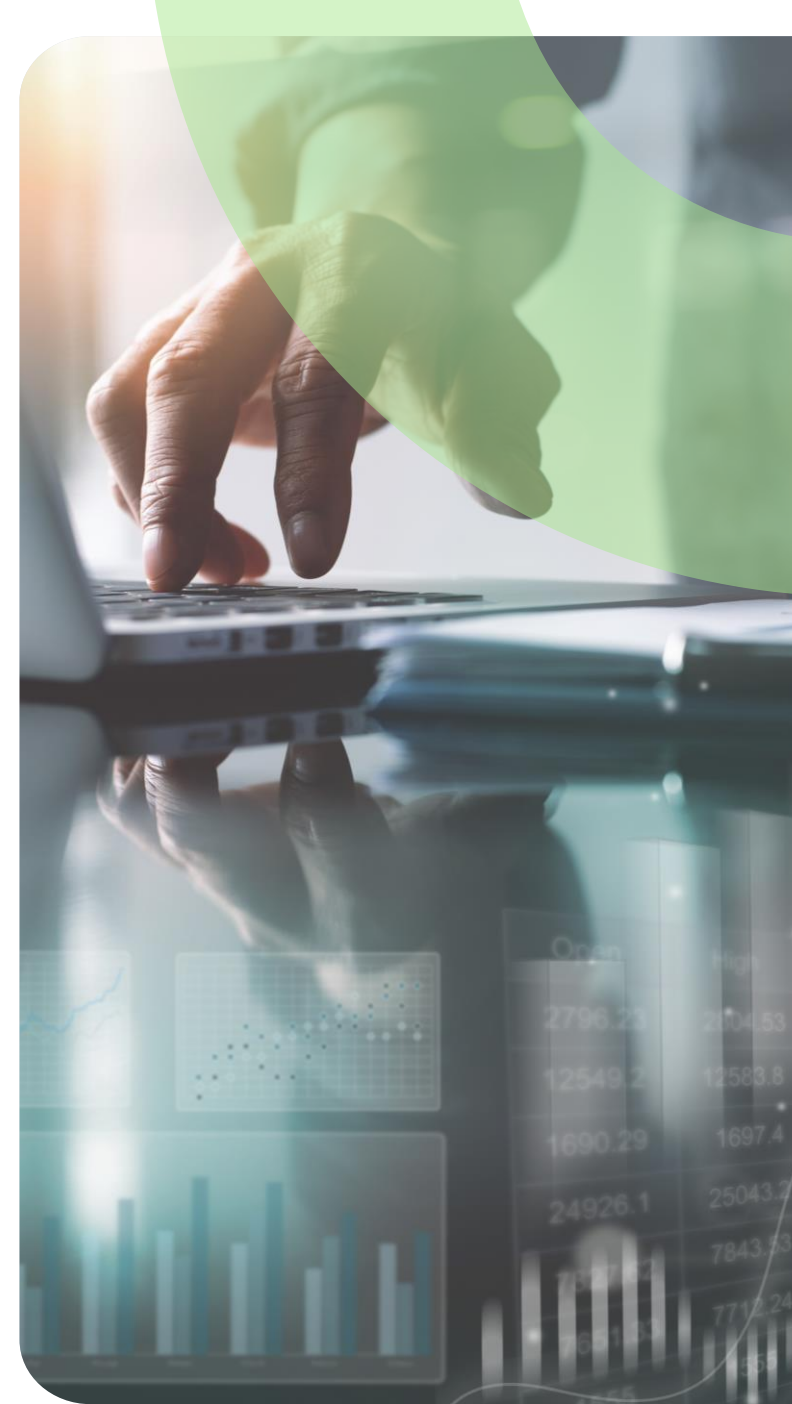
Correlation does not automatically indicate causation

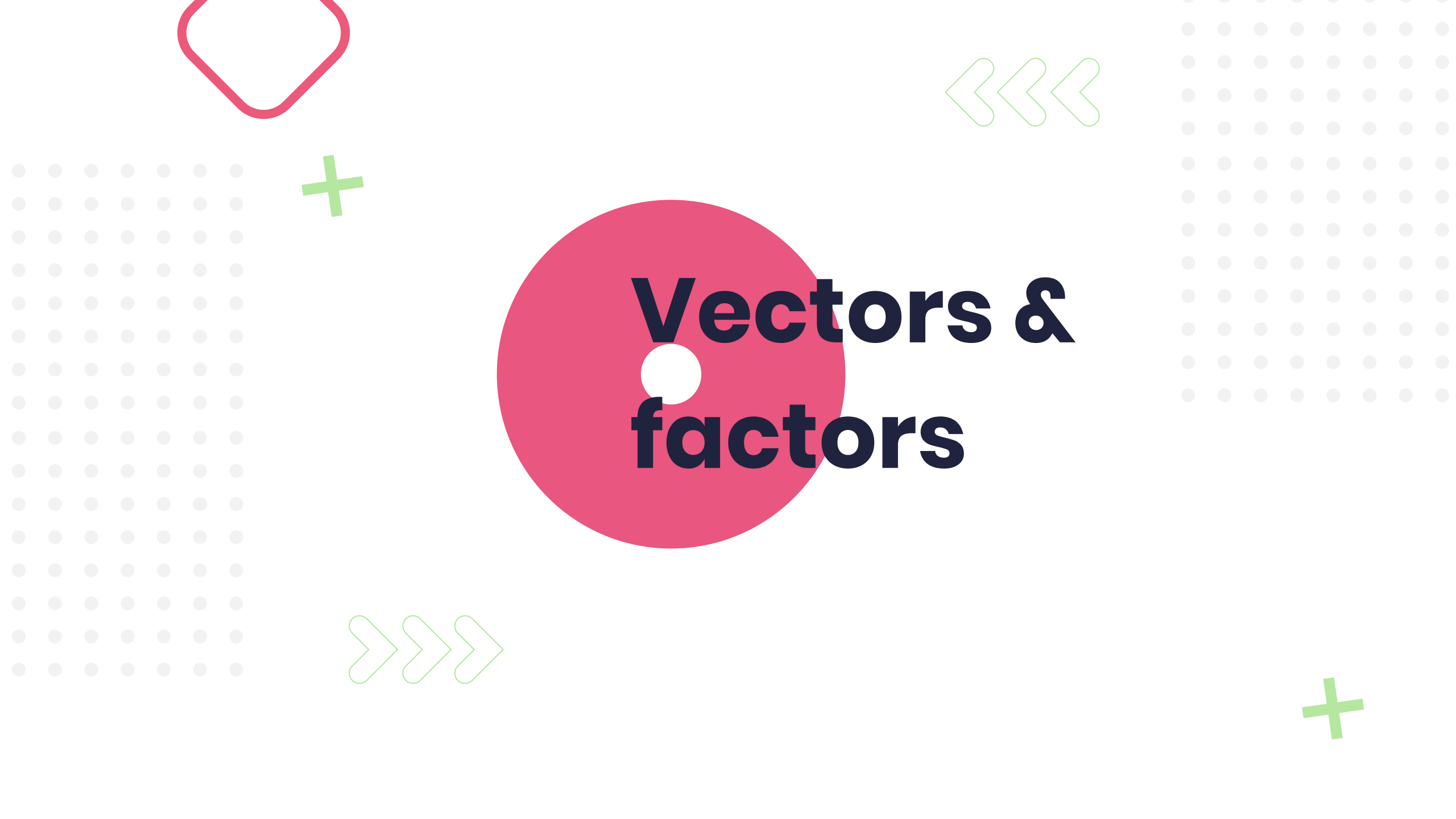




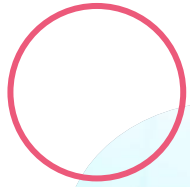
Correlation in R

- Pearson's correlation coefficient:
 - `cor(x, y)`
 - `cor.test()`





Vectors & factors



Basic commands in R

- Functions
 - `funcname(input1, input2)`
- Vector creation
 - `x <- c(1, 2, 3)`
 - `x = c(1, 2, 3)`

Basics in R

- Length of set
 - `length(x)`
- List of objects
 - `ls()`
- Remove object
 - `rm()`





Challenge >>

- Input data manually
- Create scatterplot of data

#exploredata